Math Jenga:

One-Step Equations

1. $x + 7 = 15 \rightarrow x = 8$
2. $y - 4 = 9 \rightarrow y = 13$
3. $5x = 35 \rightarrow x = 7$
4. $n/3 = 6 \rightarrow n = 18$
5. $x - 12 = -3 \rightarrow x = 9$
6. $-4x = 28 \rightarrow x = -7$
7. $y + 15 = 10 \rightarrow y = -5$
8. $n/(-2) = 9 \rightarrow n = -18$
9. $x + (-6) = 2 \rightarrow x = 8$

Two-Step Equations

10. $2x + 5 = 17 \rightarrow x = 6$
11. $3n - 4 = 11 \rightarrow n = 5$
12. $-2x + 9 = 3 \rightarrow x = 3$
13. $x/4 + 3 = 7 \rightarrow x = 16$
14. $5x - 8 = -3 \rightarrow x = 1$
15. $4 - 2x = 10 \rightarrow x = -3$
16. $3(x) + 6 = 6$... *(use $6x + 6 = 30$)* $\rightarrow x = 4$
17. $-5x - 10 = 20 \rightarrow x = -6$
18. $x/(-3) + 2 = 5 \rightarrow x = -9$

Distributive Property

19. Simplify: $3(x + 4) \rightarrow 3x + 12$
20. Simplify: $-2(x - 5) \rightarrow -2x + 10$
21. Solve: $2(x + 3) = 14 \rightarrow x = 4$
22. Simplify: $4(2x - 1) \rightarrow 8x - 4$
23. Solve: $3(x - 2) = 9 \rightarrow x = 5$
24. Simplify: $-3(2x + 5) \rightarrow -6x - 15$
25. Solve: $5(x + 1) = 20 \rightarrow x = 3$
26. Simplify: $6(x - 4) + 2x \rightarrow 8x - 24$
27. Solve: $-2(x - 3) = 10 \rightarrow x = -2$

Slope (given two points or a graph description)

28. Find slope between (2, 3) and (4, 7) → $m = 2$
29. Find slope between (0, 0) and (5, 10) → $m = 2$
30. Find slope between (-1, 4) and (3, -4) → $m = -2$
31. Find slope between (2, 5) and (2, 9) → undefined (vertical line)
32. Find slope between (-3, 6) and (4, 6) → $m = 0$
33. Find slope between (1, 1) and (4, 10) → $m = 3$
34. Find slope between (-2, -3) and (2, 5) → $m = 2$
35. Find slope between (0, 4) and (2, 0) → $m = -2$
36. Find slope between (6, 1) and (3, 1) → $m = 0$

Slope-Intercept Form ($y = mx + b$)

37. Write the equation of a line with slope 3 and y-intercept 5 → $y = 3x + 5$

38. Identify slope and y-intercept: $y = -2x + 7 \rightarrow m = -2, b = 7$
39. Write in slope-intercept form: $y - 4 = 2(x - 1) \rightarrow y = 2x + 2$
40. Identify slope and y-intercept: $y = (1/2)x - 3 \rightarrow m = 1/2, b = -3$
41. Write the equation of a line with slope -4 passing through (0, 2) $\rightarrow y = -4x + 2$
42. Convert to slope-intercept form: $2x + y = 6 \rightarrow y = -2x + 6$
43. Write the equation with slope 0 and y-intercept $-1 \rightarrow y = -1$
44. Identify slope and y-intercept: $y = x \rightarrow m = 1, b = 0$
45. Write the equation of a line with slope 5 through (0, -3) $\rightarrow y = 5x - 3$

A few practical notes for building the blocks:

- Consider color-coding blocks by topic (e.g., a small sticker or marker dot) so you can control how evenly categories come up, or so a struggling student can be steered toward an easier category if needed
- For slope/slope-intercept/standard form questions, you may want a small reference card nearby (or allow notes) since these lean more on recall of a formula than the equation-solving ones

  Idea:

  Number the jenga

  Print out a paper for them to write the answers down while they play